

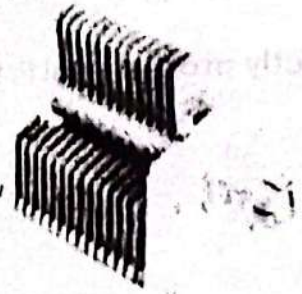
CAPACITORS:

"A Capacitor is an electronic device, which is used to store electric charge or electrical energy"

A system of two conductors separated by air or any insulating material forms a capacitor. Two conductors, isolated from one another and from their surroundings, form a capacitor.

A capacitor is said to be charged if its plates carry equal and opposite charges $+Q$ and $-Q$.

**Types of Capacitors:**

Capacitor Type	Definition	Dielectric Material	Applications
Variable Capacitor 	A capacitor whose capacitance can be changed by rotating or sliding a knob or other mechanism.	Air or mica	Tuning transmitters and receivers' signals
Film Capacitor 	A capacitor that uses a thin film of plastic or paper as the dielectric material.	Polyester, polypropylene, polystyrene, metallized paper, etc.	Power electronics devices, phase shifters, X-ray flashes and pulsed lasers

Ceramic Capacitor



A capacitor that uses a ceramic material as the dielectric material.

Ceramic

High-frequency circuits ranging from audio to radio frequency

Electrolytic Capacitor



A capacitor that uses an electrolyte as the dielectric material.

Aluminum oxide or tantalum pent oxide

DC power supply circuits, filters, snubbers and fans.

PARALLEL PLATE CAPACITORS:

A Parallel plate capacitor consists of two conducting plates of same dimensions. These plates are placed parallel to each other. Space between the plates is filled with air or any insulating material (dielectric). One plate is connected to positive terminal while the other is connected to negative terminal of power supply. The plate connected to positive terminal acquires positive charge and the other plate connected to negative terminal acquires equal negative charge. The charges are stored between the plates of capacitor due to attraction.

Experimentally it has been found that, Electric charge stored on any one of the plates of a capacitor is directly proportional to the potential difference between the plates. i.e

$$Q \propto V$$

$$Q = (\text{constant}) V$$

$$Q = CV$$

Where C = capacitance of capacitor

Capacitance of A Capacitor:

Capacitance is the measurement of how much charge is stored in a capacitor. The ability of a capacitor to store electric charge between its plates is its capacity or capacitance

OR

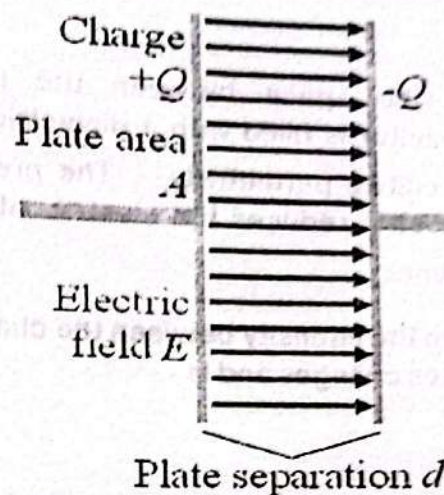
The ratio of electric charge stored on any one of the plates of capacitor to potential difference between the plates."

Mathematically

CAPACITANCE OF A PARALLEL PLATE CAPACITOR:

Consider a parallel plate capacitor as shown below

- Let
- The area of each plate = A
 - The separation between plates = d
 - Medium = air
 - Surface density of charge on each plate = σ
 - The electric field intensity between the plates of capacitor is given by



$$E = \frac{\sigma}{\epsilon_0}$$

Potential difference between the plates of capacitor can be calculated by the following relation

$$V = Ed$$

Putting the value of "E"

$$V = \frac{\sigma}{\epsilon_0} \times d$$

But $\sigma = \frac{q}{A} = \frac{Q}{A}$

$$V = \frac{Qd}{A\epsilon_0}$$

$$A\epsilon_0 V = Qd$$

We know that,

The electric charge stored on any one of plate of capacitor is $Q = CV$

$$A\epsilon_0 V = CVd$$

$$Cd = A\epsilon_0$$

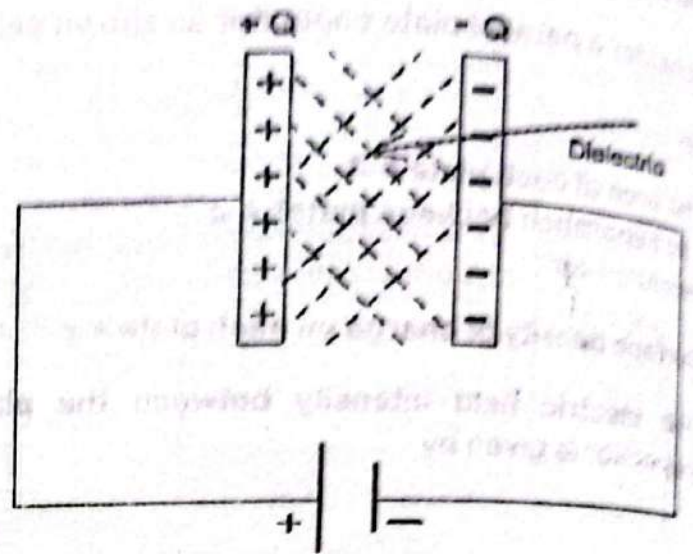
$$C = \frac{A\epsilon_0}{d}$$

This is the expression for capacitance of a parallel plate capacitor when air is the medium between the plates.

WHEN DIELECTRIC IS COMPLETELY FILLED BETWEEN THE PLATES:

Let the space between the plates of capacitor is filled with a dielectric medium of relative permittivity ϵ_r . The presence of dielectric reduces the electric intensity by ϵ_r -times

Then the intensity between the charged plates changes and is



$$E' = \frac{\sigma}{\epsilon_0 \epsilon_r}$$

$$E' = \frac{q/A}{\epsilon_0 \epsilon_r}$$

$$E' = \frac{q}{A \epsilon_0 \epsilon_r}$$

The potential difference between the plates is given by

$$V' = E'd$$

$$V' = \frac{qd}{A \epsilon_0 \epsilon_r}$$

$$\frac{q}{V'} = \frac{A \epsilon_0 \epsilon_r}{d}$$

$$\text{As } q V' = C'$$

$$C' = \frac{A \epsilon_0 \epsilon_r}{d}$$

Relative Permittivity:

Relative permittivity (ϵ_r) or dielectric constant of a medium is defined as the ratio between capacitance of a capacitor with dielectric to the capacitance without dielectric.

Value of ϵ_r lies between 1 to 10 for different materials for air = 1

$$\epsilon_r = C'/C$$

Effect of Dielectric Polarization

Dielectric is an insulator. Its molecules or atoms are non-polarized. But when it is inserted between charged capacitors, its molecules becomes distorted or polarized. These distorted molecules draw the charge to the plates inwards. Thus "E" is reduced and more charge can accumulate on plate and the capacitance of a capacitor increases.

Advantages of Dielectric

- Plates can be brought closer, increasing capacitance.
- High voltage can apply and more charge can store.

Factor Effecting the Capacitance Of A Capacitor

- Area of plates.
- Distance between the plates.
- Dielectric between the plates.

Limit of the Capacitance

There is a limit to the ability of the capacitor to store charge because as its potential ratio to a breaking value the charges store start leaking to the atmosphere.

Units of Capacitance: In S.I. system unit of capacitance is Coulomb /Volt which is generally known as farad.

Farad is a large unit therefore in general practice we use small units.

μF (microfarad) = $1 \mu\text{F} = 1 \times 10^{-6} \text{ F}$

nF (nanofarad) = $1 \text{ nF} = 1 \times 10^{-9} \text{ F}$

pF (Pico farad) = $1 \text{ pF} = 1 \times 10^{-12} \text{ F}$

Chapter

9

Combination of Capacitors

COMBINATION OF CAPACITORS:

There are two ways to connect multiple capacitors in electrical circuits.

- Series Combination
- Parallel Combination

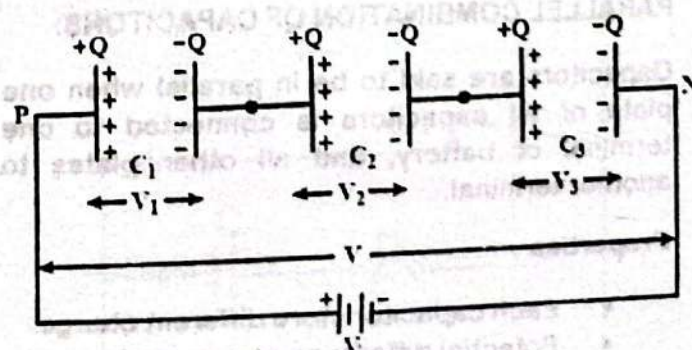
SERIES COMBINATION OF CAPACITORS:

Capacitors are said to be in series when only one terminal is common between them. They are arranged in plate to plate in series manner

As shown in the figure.

Properties

- Each capacitor store same charge q.
- Potential difference across each capacitor is different.



Series combination of capacitors

Derivation

As shown capacitances C_1 , C_2 and C_3 are connected in series. As q is constant due to single path of electricity flow.

$$V_1 = \frac{q}{C_1}$$

$$V_2 = \frac{q}{C_2}$$

$$V_3 = \frac{q}{C_3}$$

If these capacitors are replaced by a single equivalent capacitor C_e , which will store same charge q , then

$$V = \frac{q}{C_e}$$

Here,

$$V = V_1 + V_2 + V_3$$

Therefore

$$\frac{q}{C_e} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$$

Or

$$\frac{q}{C_e} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$$

$$\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Therefore we can say that when Capacitors are connected into series the reciprocal of the equivalent capacitance is equal to the sum of reciprocal of individual's capacitances.

Conclusion

- The equivalent capacitance is less than the smallest individual capacitance.
- Capacitors in series may be regarded as single capacitor of greater plate separation.

Application

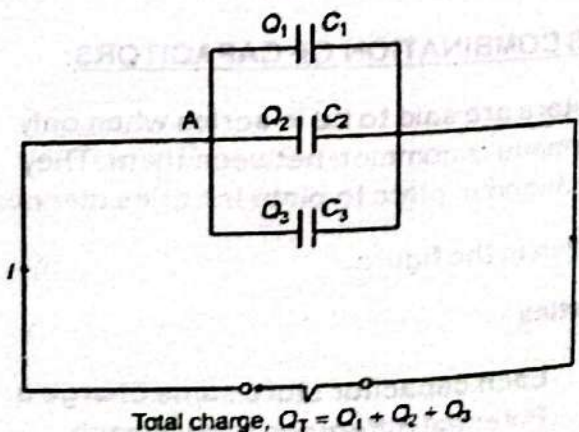
Series connection is used when it is desired to divide a high voltage among several capacitors. If any one of which cannot sustain electric voltage.

PARALLEL COMBINATION OF CAPACITORS:

Capacitors are said to be in parallel when one plate of all capacitors is connected to one terminal of battery, and all other plates to another terminal.

Properties

- Each capacitor store different charge.
- Potential difference across each capacitor is same due to same terminal point for each capacitor.



Derivation

As shown capacitances C_1 , C_2 and C_3 are connected in parallel, now

$$Q_1 = C_1 V$$

$$Q_2 = C_2 V$$

$$Q_3 = C_3 V$$

If these capacitors are replaced by a single equivalent capacitor C_e which will store total charge Q_T , then

$$Q_T = C_e V$$

But

$$Q_T = Q_1 + Q_2 + Q_3$$

$$C_e V = C_1 V + C_2 V + C_3 V$$

Or

$$C_e = C_1 + C_2 + C_3$$

Therefore we can say that when Capacitors are connected into parallel, the equivalent capacitance is equal to the sum of individual's capacitances.

Conclusion

- The equivalent capacitance is greater than the greatest individual capacitance.
- Capacitors in parallel may be regarded as single capacitor of greater plate area.

Application

Parallel connection is used when large capacitance is required at a moderate or low potential.

Chapter

9

Charging and Discharging of Capacitors

CHARGING AND DISCHARGING OF CAPACITORS THROUGH RESISTORS:

Consider a circuit having a capacitance C and a resistance R which are joined in series with a battery of e.m.f " ϵ " through a Morse key K , as shown in the figure.

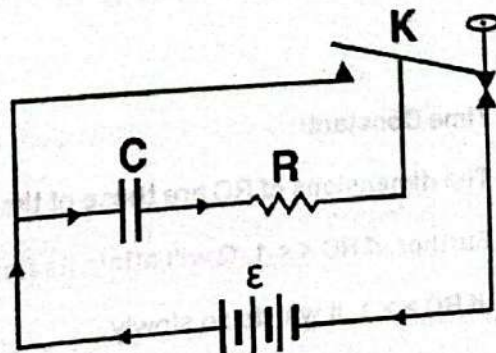
Charging of a Capacitor:

When the key is pressed, the capacitor begins to store charge. If at any time during charging, I is the current through the circuit and Q is the charge on the capacitor, then

The potential difference across resistor = IR ,

The potential difference between the plates of the capacitor = Q/C

Since the sum of both these potentials is equal to ϵ ,



FOR CHARGING:

If $\tau = t$, then, from equation,

$$Q = Q_0(1 - e^{-1})$$

or

$$\begin{aligned} Q &= Q_0 \left(1 - \frac{1}{2.718}\right) \\ &= Q_0(1 - 0.368) = 0.632Q_0 \\ &= 63.2\% \text{ of } Q_0 \end{aligned}$$

The time constant of a CR circuit is thus the time during which the charge on the capacitor becomes 0.632 (approx., 2/3) of its maximum value.

For the charge on the capacitor to attain its maximum value (Q_0), i.e., for $Q = Q_0$,

$$e^{-t/RC} = 0 \text{ or } t = \infty$$

Thus, theoretically, the charge on the capacitor will attain its maximum value only after infinite time.

FOR DISCHARGING:

If $\tau = t$, then, from equation,

$$Q = Q_0 e^{-1}$$

Or

$$\begin{aligned} Q &= Q_0/e \\ Q &= 0.368Q_0 \\ Q &= 36.8\% \text{ of } Q_0 \end{aligned}$$

The time constant of a CR circuit is thus also the time during which the charge on the capacitor falls from its maximum value to 0.368 (approx., 1/3) of its maximum value.

Thus, the charge on the capacitor will become zero only after infinite time.

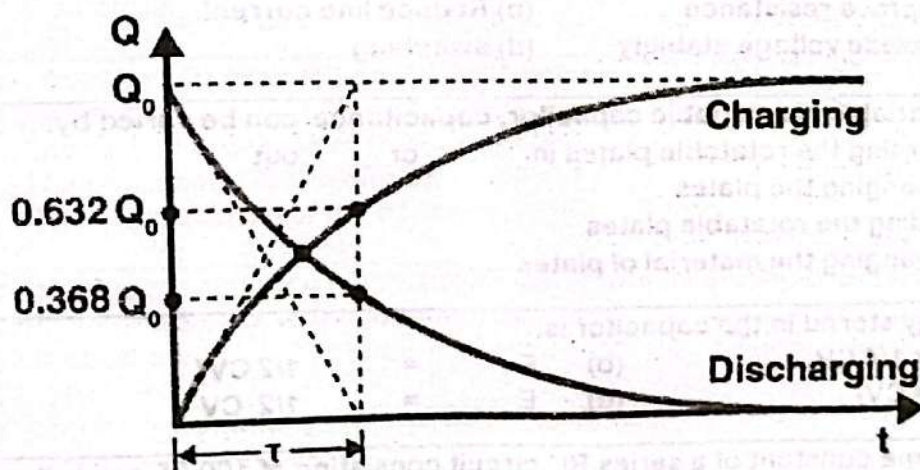


DIAGRAM DEPICTING CHARGING AND DISCHARGING OF CAPACITOR

Energy Stored in Capacitors:

Energy stored in a capacitor is electrical potential energy, and it is thus related to the charge q and voltage V on the capacitor.

If the capacitance of a conductor is C , then it is initially uncharged and it acquires a potential difference V when connected to a battery. If $+Q$ and $-Q$ charge are deposited on the plates, then the average voltage on the capacitor during the charging is $V/2$.

$$\text{Energy} = U = \frac{QV}{2}$$

Substituting

$$Q = CV$$

In the equation above, we get

$$E = U = \frac{1}{2} CV^2$$

The energy stored in a capacitor can be expressed in three ways:

$$E_{\text{cap}} = \frac{QV}{2} = \frac{CV^2}{2} = \frac{Q^2}{2C}$$

Chapter

9

Multiple Choice Questions

MCQ'S		ANS.
1. The capacitance of a capacitor is NOT influenced by (a) Plate thickness (b) Plate area (c) Plate separation (d) Nature of the dielectric		a
2. What is the value of capacitance of a capacitor which has a voltage of 4V and has 16C of charge? (a) 2F (b) 4F (c) 6F (d) 8F		b
3. Capacitors are used in electric power supply system to: (a) Improve resistance (b) Reduce line current (c) Provide voltage stability (d) switching		c
4. In a variable capacitor, capacitance can be varied by: (a) Turning the rotatable plates in or out (b) Changing the plates (c) Sliding the rotatable plates (d) Changing the material of plates		a
5. Energy stored in the capacitor is: (a) $E = 1/4 CV$ (b) $E = 1/2 CV^2$ (c) $E = CV^2$ (d) $E = 1/2 CV$		b
6. The time constant of a series RC circuit consisting of 100J.F capacitor in series with a 100Ω resistor is. (a) 0.1s (b) 0.1 ms (c) 0.01s (d) 0.01 ms		c
7. The charging of a capacitor through a resistance follows (a) linear law (b) square law (c) exponential law (d) none of the above		c
8. When the total charge in a capacitor is doubled, the energy stored (a) remains the same (b) is doubled (c) is halved (d) is quadrupled		d

9. The capacitance C is charged through a resistor R . The time constant of the charging circuit is given by
- (a) C/R (b) $1/RC$ (c) RC (d) R/C

10. Capacitor blocks:
- (a) alternating current
(b) direct current
(c) both alternating and direct current
(d) neither alternating nor direct current

***** Exams Practice MCQs *****

1. What type of dielectric material is commonly used in capacitors to increase their capacitance?

- a) Non-conductive.
b) Conductive.
c) Magnetic.
d) Metallic.

2. How many conductors does the capacitor consist of?

- a) One
b) Two
c) Three
d) Four

3. What is the standard unit of electric charge?

- a) Watts
b) Henry
c) Farads
d) Coulombs

4. The time constant (τ) in an RC circuit determines:

- a) The energy stored in the capacitor.
b) The rate of charging of the capacitor.
c) The resistance of the circuit.
d) The capacitance of the capacitor.

5. If you double the capacitance (C) in an RC circuit while keeping the resistance (R) constant, what happens to the time constant (τ)?

- a) It doubles.
b) It halves.
c) It remains unchanged.
d) It becomes zero.

6. When a capacitor is fully charged in an RC circuit, which of the following statements is true?

- a) Voltage across the capacitor is zero.
b) Current through the resistor is zero.
c) Voltage across the capacitor is at its maximum.
d) Voltage across the resistor is at its maximum.

7. During the discharging of a capacitor, what happens to the voltage across the capacitor over time?

- a) It increases exponentially.
b) It decreases exponentially.
c) It remains constant.
d) It oscillates.

Physics XI

8. In an RC circuit, if the resistance (R) is increased while the capacitance (C) is kept constant, what happens to the time constant (τ)?

- a) It decreases.
- b) It increases.
- c) It remains the same.
- d) It becomes zero.

9. What is the primary function of a parallel plate capacitor?

- a) To store energy in the form of magnetic fields.
- b) To store energy in the form of electric fields.
- c) To regulate current flow in a circuit.
- d) To amplify electrical signals

10. In a parallel plate capacitor, if you increase the distance between the plates while keeping the charge and plate area constant, what happens to the capacitance?

- a) It increases.
- b) It decreases.
- c) It remains the same.
- d) It becomes zero.

11. If the voltage across a parallel plate capacitor is doubled while keeping the charge constant, what happens to the electric field strength between the plates?

- a) It doubles.
- b) It halves.
- c) It remains the same.
- d) It becomes zero.

12. When a dielectric material is inserted between the plates of a parallel plate capacitor, what effect does it have on the capacitance?

- a) Increases the capacitance.
- b) Decreases the capacitance.
- c) Has no effect on capacitance.
- d) Reverses the polarity of the plates.

13. When capacitors are connected in series, what happens to their total capacitance compared to individual capacitances?

- a) It increases.
- b) It decreases.
- c) It remains the same.
- d) It becomes zero.

14. If three capacitor each of $10\ \mu\text{F}$ capacitors are connected in parallel the, net capacitance is

- a) $20\ \mu\text{F}$
- b) $30\ \mu\text{F}$
- c) $50\ \mu\text{F}$
- d) $40\ \mu\text{F}$

15. If the voltage (V) across a capacitor is doubled while keeping the capacitance (C) constant, how does the energy stored in the capacitor change?

- a) It doubles.
- b) It quadruples.
- c) It halves.
- d) It remains the same.

16. What happens to the energy stored in a capacitor if the distance between the plates is decreased while keeping the voltage constant?

- a) It increases.
- b) It decreases.
- c) It remains the same.
- d) It becomes zero.

17. The capacitance of a capacitor is directly proportional to:

- a) The voltage across it.
- b) The charge stored in it.
- c) The area of its plates.
- d) The resistance in the circuit.

18. What happens to the electric field strength between the plates of a capacitor when a dielectric material is inserted?

- a) It decreases.
- b) It increases.
- c) It remains the same.
- d) It becomes zero.

19. What will be the charge on each capacitor having values $20\ \mu\text{F}$, $10\ \mu\text{F}$ and $30\ \mu\text{F}$ are connected in series.

- a) low capacitor has low charge.
- b) each having same charge.
- c) high capacitor has high charge.
- d) low capacitor has no charge

20. In an RC circuit, the 63% charging time is often referred to as:

- a) The critical time
- b) The time constant
- c) The half-time
- d) The peak time

Chapter**9****Chapter Formulae & Numericals****Charge of Capacitor**

$$Q = CV$$

Capacitance of parallel plate capacitor

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance of parallel plate capacitor with dielectric other than air

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

Parallel Combination of Capacitor

$$C_{eqv} = C_1 + C_2 + C_3 \dots$$

Series combination of Capacitor

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \dots$$

Charging of Capacitor

$$Q = Q_0(1 - e^{-t/R})$$

Discharging of Capacitor

$$Q = Q_0 e^{-t/R}$$

Time Constant of Capacitor

$$\tau = RC$$

Energy stored in a Capacitor

$$U = \frac{1}{2} CV^2$$

$$V_R + V_C = \epsilon$$

$$IR + Q/C = \epsilon$$

As the current stops flowing when the capacitor is fully charged,
When $Q = Q_0$ (the maximum value of the charge on the capacitor), $I = 0$
Then Final Equation is found out to be:

$$Q = Q_0(1 - e^{-t/RC})$$

or

$$Q = Q_0(1 - e^{-t/\tau})$$

$$\tau = RC$$

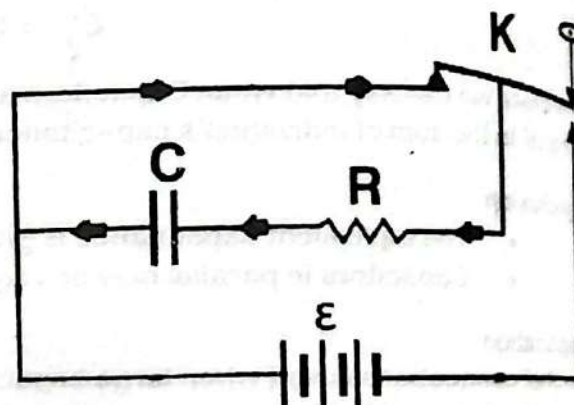
Where

It gives us the value of charge on the capacitor at any time during charging

Discharging of a Capacitor

When the key K is released [Figure], the circuit is broken without introducing any additional resistance. The battery is now out of the circuit, and the capacitor will discharge itself through R.

If I is the current at any time during discharge, then putting $\epsilon = 0$



$$IR + Q/C = \epsilon$$

if $\epsilon = 0$

$$IR + Q/C = 0$$

The Final Equation we get will be:

$$Q = Q_0 e^{-t/RC} = Q_0 e^{-t/\tau}$$

Where

$$\tau = RC$$

Time Constant:

The dimensions of RC are those of time.

Further, if $RC \ll 1$, Q will attain its final value rapidly

If $RC \gg 1$, it will do so slowly

Thus RC defines the rate of charging and discharging of capacitors